

# Fight Drones with Drones: Detecting Aerial Perimeter Intrusion using Drone-mounted Microphones

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**Abstract.** While in recent years there has been a growing interest in detecting drones via their characteristic sound signature, especially through ground-based sensory capture, the usage of drones to detect the presence of other drones has shown little support and, even more so, the usage of drones to detect perimeter violation by other drones. In this paper, we are the first to propose, to the best of our knowledge, a framework for deploying a drone swarm as an adaptive acoustic detection barrier. One application of this solution is to detect opponents' drone-perimeter violation of critical infrastructure such as airports, energy plants, and national borders, to cite a few. Mounting microphones on patrolling defender drones turns a static detection footprint into an adaptive coverage layer. In detail, starting from an acoustic model calibrated on real DJI flight data for the Mavic Air 2 and Mini Pro 3, we show that monitoring a  $500 \times 200$  m boundary with a static grid without any gap requires over 400 surveillance drones. Whereas, leveraging mobility and adopting an optimal patrol speed, we reduce the number of needed drones by more than 90%, while still having a 99% detection probability to detect the intruder. We detail the parameters that influence the detection and show that, among others, a 6dB reduction in intruder acoustic signature doubles the required fleet, while any improvement in ego-noise cancellation yields defense a disproportionate advantage. Finally, we carry out a complete analysis covering energy budgets, battery-swap logistics, and collision-safe spacing, providing sizing tables directly applicable to real-life deployments such as the protection of oil refineries, airports, or harbors.

**Keywords:** Drone swarm · Acoustic detection · Perimeter security · Critical Infrastructures Protection · Monte Carlo simulation · Ego-noise cancellation

## 1 Introduction

Unauthorized drone incursions are an escalating threat to the security of airports, energy infrastructure, national borders, and military installations. The U.S. Federal Aviation Administration receives over 100 drone sighting reports near airports every month [11], with restricted-airspace incursions surging 26% year-on-year in early 2025 [30]. The December 2018 Gatwick Airport closure, which grounded roughly 1000 flights and cost airlines an estimated €55.8 million over 33 hours [9, 37], demonstrated how a single low-cost drone can paralyze critical infrastructure. Similar incidents have since disrupted Frankfurt Airport on multiple occasions [12] and forced ground stops at several U.S. airports [32]. The EASA estimates that a 30-minute runway shutdown at a major European hub generates €325 000–514 000 in delay costs alone [9].

The threat extends well beyond aviation. The 2019 drone strike on the Abqaiq and Khurais oil facilities in Saudi Arabia temporarily halved the country’s crude output [21]. The Russia-Ukraine conflict has since normalized drone warfare at an industrial scale, with both sides deploying thousands of one-way attack and loitering munitions per month against refineries, airfields, and civilian infrastructure [7, 17, 23]. In the civilian domain, drones are routinely used for contraband smuggling into prisons and for surveillance of sensitive sites [6]. Consumer multirotors, available for a few hundred dollars and largely invisible to radar at low altitude, create a deep cost asymmetry favoring the attacker [21, 27].

No single counter-UAS sensing modality addresses the threat comprehensively [21, 27]. RF analysis fails against autonomous platforms; radar loses sensitivity against small targets near the ground [3, 20]; and electro-optical/infrared cameras degrade in darkness, fog, or rain. Passive acoustic detection fills a complementary niche: every multirotor radiates broadband propeller noise as a by-product of flight, detectable without emitting interrogation energy and largely unaffected by weather [4, 28]. However, existing ground-mounted arrays offer only a static, fixed-footprint detection layer. Most existing drone authentication approaches [14] remain limited to identity verification and cannot determine whether a UAV has actually traversed a protected boundary.

Mounting microphones on patrolling defender drones would transform this static footprint into an adaptive perimeter curtain, but it introduces what we term the Defender’s Dilemma: forward flight raises the onboard ego-noise floor, shrinking the detection range the patrol is designed to extend. On a planar boundary, this penalty propagates quadratically into fleet sizing: halving the detection range quadruples the number of defenders required for gap-free coverage. No prior work has quantified this trade-off or optimized for it in an operational setting.

**Contributions.** This paper introduces the following contributions:

- We formalize the speed–detection trade-off for airborne acoustic sensing and show, using a model calibrated from real DJI flight data, that a static gap-free grid on a  $500 \times 200$  m boundary requires  $N_{\text{cov}} \approx 400$  defenders, whereas mobilizing a fleet roughly ten times smaller at the optimal patrol speed of 35.4 km/h achieves a 99% detection through temporal coverage integration.

- We derive a quadratic fleet-sizing law for planar perimeter defense and validate it across four intruder archetypes, showing that a 6 dB reduction in the attacker’s acoustic signature doubles the required fleet.
- We characterize the adversarial equilibrium between attacker stealth and transit speed across four drone archetypes, demonstrating the existence of an archetype-specific optimal attacker speed that minimizes the probability of detection.
- We present a complete operational analysis: energy budgets, battery-swap schedules, and spacing for collision avoidance. The results provide concrete sizing tables for deployments in airports, border segments, and energy facilities.

**Organization.** Section 2 surveys background material. Section 3 presents the system model. Section 4 describes the empirical calibration. Section 5 introduces the simulation engine. Section 6 reports the experiments. Section 7 derives design guidelines and discusses limitations. Section 8 surveys related work. Section 9 concludes.

## 2 Background

### 2.1 Counter-UAS Sensing Modalities

Small consumer multirotors are elusive targets for air-defense sensors: their radar cross-section is comparable to that of birds, and they routinely operate below the coverage floor of primary surveillance radars [3, 20]. Modern counter-UAS architectures therefore fuse several complementary modalities [21, 27]. Radio-frequency analysis intercepts the control link but is blind to autonomous platforms [21, 27]. In addition, network-traffic-based methods detect drones by analyzing controller–UAV communication patterns [25]. However, such approaches depend on active communication links and fail when drones operate autonomously. Radar is effective at range but loses sensitivity against lightweight targets at low altitude [10]. Electro-optical and infrared cameras offer high resolution in clear weather but degrade in fog, rain, or at night [22]. Acoustic sensing [15] fills a distinct gap: it is passive, omnidirectional, and operational in most weather conditions, but at the cost of a limited detection range—typically 300–500 m for ground-mounted arrays [4].

### 2.2 Acoustic Signatures of Multirotors

The acoustic signature of a multirotor is dominated by its blade-pass frequency  $f_{bp} = n_b \cdot \Omega / 60$  and its harmonics, where  $n_b$  is the blade count and  $\Omega$  the rotational speed in rpm [13, 31]. Forward flight adds broadband noise from blade-loading variations and turbulent inflow. Torija and Clark [31] measure how the resulting noise footprint scales with rotor speed and airspeed, providing multi-UAV audio references exploitable for ego-noise cancellation. The DREGON dataset [28] supplies spatially annotated multi-UAV recordings suitable for training and validation.

Table 1: Principal notation.

| Symbol                           | Definition                   | Symbol                   | Definition                    |
|----------------------------------|------------------------------|--------------------------|-------------------------------|
| $\mathcal{W}$                    | Defended boundary wall       | $L_{\text{ego}}^{(i)}$   | Ego-noise floor (dBSPL)       |
| $N$                              | Total number of defenders    | $\text{SNR}_{\text{th}}$ | Minimum SNR threshold         |
| $u_{\text{nt}} / u_{\text{def}}$ | Intruder / defender speed    | $R_{\text{det}}$         | Effective detection range     |
| $v^*$                            | Optimal patrol speed         | $N_{\text{cov}}$         | Gap-free minimal fleet size   |
| $u_{\text{blind}}$               | Acoustic blindness threshold | $P_{\text{det}}$         | Prob. of successful detection |
| $L_{\text{src}}(v)$              | Source level (dBSPL at 1 m)  | $\alpha_{\text{atm}}$    | Acoustic absorption (dB/m)    |

### 2.3 Ego-Noise in Airborne Acoustic Sensing

Mounting a microphone on a flying drone introduces a fundamental challenge: the platform’s own rotor noise can exceed the target signal by 20 dB or more. Cabrera-Ponce et al. [5] demonstrate drone-to-drone detection using a multi-microphone array carried by a patrolling UAV. Wang and Cavallaro [36] combine a suspended microphone with spectral subtraction to suppress ego-noise during species monitoring, reporting 10–15 dB SNR gains. Search-and-rescue systems [29] apply enhanced beamforming to drone-borne arrays but assume hovering or slow flight. None of these works studies how ego-noise scales with speed across a multi-defender swarm.

### 2.4 Multi-Agent Coverage Control

A central question in this work is how to position and move a swarm of defenders so that their detection footprints collectively seal a defended perimeter. Cortés et al. [8] formalized coverage control for mobile sensing networks using Voronoi partitions and Lloyd’s algorithm. Subsequent extensions incorporate potential-field methods and ergodic coverage [18, 19]. These formulations assume that sensing quality degrades with distance alone. In our setting, a defender that moves faster covers more ground but simultaneously raises its own noise floor, shrinking its detection range. Zorbas et al. [40] study optimal drone placement for target surveillance, minimizing fleet size and energy. Their formulation captures a related trade-off but operates on static placements with geometric optical coverage and does not model an adversary or self-generated noise.

## 3 System Model and Assumptions

This section formalizes the boundary geometry, the threat model, the acoustic sensing capabilities, and the mobility dynamics of the drone swarm. Table 1 summarizes the principal notation.

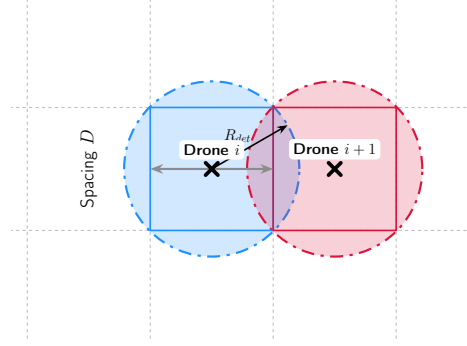


Fig. 1: Boundary wall geometry: defenders are arranged on the perimeter plane  $\mathcal{W}$  with inter-drone spacing  $D$  and individual detection radius  $R_{\text{det}}$ . The intruder approaches orthogonally.

### 3.1 Boundary Geometry and Spatial Model

We consider a critical infrastructure perimeter protected by a cooperative swarm of  $N$  multirotor drones equipped with microphone payloads. The defended boundary is modeled as a planar wall  $\mathcal{W} = [0, W_y] \times [z_{\min}, W_z]$ , as illustrated in Figure 1. The  $y$ -axis represents the lateral extent of the perimeter and  $z$  denotes altitude, bounded below by a minimum safe patrol altitude  $z_{\min}$ . The intruder's approach axis is orthogonal to  $\mathcal{W}$ . The swarm dynamically adjusts its positions across  $\mathcal{W}$  to form an acoustic detection curtain. This geometry directly captures the operational reality of airport fence lines, pipeline corridors, and border segments.

### 3.2 Threat Model and Attacker Dynamics

**Attacker objective.** The adversary is an unauthorized multirotor attempting to covertly traverse the defended boundary  $\mathcal{W}$  without triggering an acoustic alert. The intruder enters the patrol zone from a uniformly random position on the outer face, flies at constant speed  $v_{\text{int}}$  along a trajectory directed toward the protected interior, and succeeds if it crosses  $\mathcal{W}$  without detection. The adversary balances transit time against acoustic stealth: a higher  $v_{\text{int}}$  minimizes exposure time within the defenders' range but increases the broadband rotor noise  $L_{\text{src}}(v_{\text{int}})$ .

**Attacker knowledge model.** We adopt a zero-knowledge attacker as the baseline: the adversary does not know whether a protective drone wall is deployed, where it is located, the number of defenders  $N$ , their spatial formation, or the exact value of the acoustic blindness threshold  $v_{\text{blind}}$ . This is a conservative assumption for the defender, as a more informed attacker could time a crossing at a predictable coverage gap.

**Scope: detection, not neutralization.** This work addresses acoustic detection of intrusion events. The downstream response, e.g., jamming [33, 35], is orthogonal to the detection problem and is not modeled. We note that a deliberate adversarial strategy could involve saturating the perimeter with multiple low-cost acoustic emitters to mask a true intrusion amid repeated false positives.

Mitigating such saturation attacks is also considered orthogonal to this work. Adversarial games involving dynamic attacker strategies, under a stronger threat model, are therefore left for future work.

### 3.3 Acoustic Sensing and Ego-Noise Model

**Detection condition.** Defender  $i$  raises a detection alert at time  $t$  when its received signal-to-noise ratio exceeds a threshold  $\text{SNR}_{\text{th}}$ :

$$\text{SNR}_i(t) = L_{\text{rx}}^{(i)}(t) - L_{\text{ego}}^{(i)}(t) \geq \text{SNR}_{\text{th}}, \quad (1)$$

where  $L_{\text{rx}}^{(i)}$  is the intruder's acoustic power received at drone  $i$  and  $L_{\text{ego}}^{(i)}$  is the ego-noise floor after spectral cancellation. We use  $\text{SNR}_{\text{th}} = 3$  dB as the default operating point, corresponding to the knee of the empirical ROC curve (Appendix B).

**Conservative boundary assumption.** A detection event is scored exclusively if it occurs before the intruder crosses  $\mathcal{W}$ . Any fleet sizing derived from Eq. (6) is therefore a lower bound on achievable detection reliability.

### 3.4 Acoustic Emission Model

**Source level.** The broadband noise radiated by a multicopter at flight speed  $v$  (km/h) is:

$$L_{\text{src}}(v) = L_0 + g \cdot \min(v, v_{\text{pl}}) \quad [\text{dB SPL at 1 m}], \quad (2)$$

where  $L_0$  is the hovering source level,  $g$  the linear speed gain (dB h/km), and  $v_{\text{pl}}$  the plateau speed above which aerodynamic flow separation limits further noise growth. Parameters for the four attacker archetypes are given in Table 2.

**Propagation.** Under free-field conditions, the received power at range  $d$  is:

$$L_{\text{rx}}(d, v) = L_{\text{src}}(v) - 20 \log_{10}(d) - \alpha_{\text{atm}} \cdot d, \quad (3)$$

where  $\alpha_{\text{atm}} = 0.005$  dB/m is an empirical broadband absorption coefficient for moderate humidity [1].

**Detection range.** The maximum range  $R_{\text{det}}$  at which an intruder at speed  $v$  can still be detected is the largest  $d$  satisfying:

$$L_{\text{src}}(v) - 20 \log_{10}(d) - \alpha_{\text{atm}} d \geq L_{\text{ego}}(v_{\text{def}}) + \text{SNR}_{\text{th}}. \quad (4)$$

For the Mavic Air 2 intruder at  $v = 28.8$  km/h, with calibrated  $L_{\text{ego}} = 65.7$  dB SPL and  $\text{SNR}_{\text{th}} = 3$  dB, the resulting range is  $R_{\text{det}} = 9.1$  m.

**Ego-noise under motion.** When the defender accelerates to  $v_{\text{def}}$  at altitude  $z$ :

$$L_{\text{ego}}(v_{\text{def}}, z) = L_{\text{ego}}^{\text{hov}} + g_d \cdot \min(v_{\text{def}}, v_{\text{pl},d}) + \kappa_w \cdot \max(z - z_{\text{ref}}, 0), \quad (5)$$

where  $g_d = 0.14$  dB h/km is the defender speed-noise gain,  $v_{\text{pl},d} = 28.8$  km/h the Mini 3 Pro plateau speed, and  $\kappa_w = 0.08$  dB/m an altitude-dependent wind-noise coefficient above  $z_{\text{ref}} = 10$  m.

Table 2: Drone acoustic model parameters and detection ranges at  $v = 28.8$  km/h with  $\text{SNR}_{\text{th}} = 3$  dB and spectral cancellation ( $L_{\text{ego}} = 65.7$  dBSPL).

| Platform          | $L_0$<br>(dBSPL) | $g$<br>(dB h/km) | $v_{\text{pl}}$<br>(km/h) | Mass<br>(kg) | $R_{\text{det}}$<br>@28.8 km/h |
|-------------------|------------------|------------------|---------------------------|--------------|--------------------------------|
| Mavic Air 2       | 81.5             | 0.22             | 36.0                      | 0.570        | 9.1 m                          |
| Mavic 3           | 79.0             | 0.19             | 43.2                      | 0.895        | 6.2 m                          |
| FPV Racing        | 88.0             | 0.50             | 54.0                      | 0.300        | 47.3 m                         |
| Fixed-Wing VTOL   | 75.0             | 0.08             | 90.0                      | 2.500        | 2.7 m                          |
| Mini 3 Pro (def.) | 78.0             | 0.14             | 28.8                      | 0.249        | —                              |

### 3.5 Gap-Free Coverage and the Quadratic Law

The placement geometry is illustrated in Figure 1. Defenders are tiled on a regular grid in  $\mathcal{W}$  with uniform spacing  $D = 1.6 R_{\text{det}}$ , guaranteeing a 20% overlap between adjacent detection footprints and closing every straight-line corridor. The minimum fleet for this gap-free tiling is:

$$N_{\text{cov}} = \left\lceil \frac{W_y}{D} \right\rceil \times \left\lceil \frac{W_z - z_{\min}}{D} \right\rceil. \quad (6)$$

Because  $D \propto R_{\text{det}}$ , Eq. (6) yields  $N_{\text{cov}} \propto R_{\text{det}}^{-2}$ : a quadratic law. For a hovering defender against a Mavic Air 2 intruder at  $v_{\text{int}} = 15$  km/h,  $R_{\text{det}} \approx 10.1$  m and  $D = 16.2$  m, giving  $N_{\text{cov}} = 31 \times 13 = 403$  defenders for the  $500 \times 200$  m wall. This fleet guarantees 100% detection by construction, but is impractical for most operational budgets. A central result of this work (Section 6.3) is that mobilizing a fleet roughly ten times smaller at an optimal patrol speed recovers 99% detection through temporal coverage integration: each patrolling defender sweeps a wider area over the intruder’s transit time, probabilistically closing the gaps that a sparse static grid would leave open.

### 3.6 Collision Avoidance and Safety Spacing

In a dense swarm operating on the perimeter wall, maintaining collision-safe spacing between drones is a binding operational constraint. We impose a minimum separation  $d_{\min} = 3$  m between any two defenders at all times, enforced through the spring-tethered formation model of Eq. (8). For the fleet sizes and wall dimensions studied here, the gap-free grid spacing  $s = 1.6 R_{\text{det}}$  exceeds  $d_{\min}$  by a wide margin (the smallest is  $s = 14.6$  m at  $R_{\text{det}} = 9.1$  m). At very high fleet densities or under strong wind perturbations, reactive collision-avoidance maneuvers could temporarily displace defenders from their assigned slots, opening transient coverage gaps. Quantifying this interaction under extreme conditions is considered orthogonal to the present study.

## 4 Empirical Characterization

### 4.1 Hardware and Data Collection

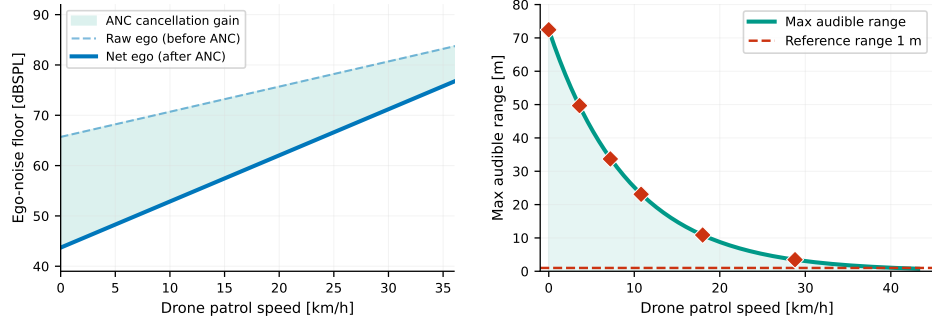
We conducted outdoor flight experiments with three DJI consumer platforms. A Mavic Air 2 (570 g, 213 mm diagonal propellers, max speed 68.4 km/h) played the intruder role. Two Mini 3 Pro units (249 g, 150 mm propellers, max speed 57.6 km/h) acted as defenders. All three drones logged telemetry at 10 Hz via the AirData UAV platform, recording GPS position (WGS-84, horizontal accuracy  $\pm 1.5$  m, vertical  $\pm 0.5$  m), altitude, three-axis speed components, throttle percentage, and battery state. The Mini 3 Pro units recorded onboard audio via RODE Wireless GO II TX with a furry windscreen to reduce direct wind noise. The Mavic Air 2 flew straight passes at commanded airspeeds from 0 to 47.2 km/h and altitudes between 0 and 15.1 m. Each defender hovered at 5.4–5.9 m altitude while continuously recording. The experiment covered 390 s of intruder flight, yielding 3904 synchronized telemetry frames at 10 Hz per drone pair.

### 4.2 Ego-Noise Measurement and Cancellation

A hovering multirotor radiates a quasi-stationary ego-noise field whose spectral structure is determined by  $f_{bp}$  and its harmonics. For the Mini 3 Pro at hover throttle,  $f_{bp} \approx 80$  Hz, producing spectral peaks at approximately 80, 160, 240, and 320 Hz.

**Friend-or-foe discrimination.** In a multi-defender swarm, neighboring defenders may enter each other’s acoustic detection range, raising the risk of false positives or, conversely, of masking a genuine intruder whose signature resembles that of a friendly platform. We assume that each defender maintains a local whitelist of its neighbors’ spectral fingerprints, obtained during a pre-mission calibration phase. Because every unit in the fleet is the same platform (Mini 3 Pro), the blade-pass harmonics are nearly identical across units, but minor manufacturing tolerances and rotor wear create per-unit spectral features that enable fingerprinting at short range [16, 24]. Additionally, each defender knows the formation geometry and can predict the expected received level from its neighbors via Eq. (3): any acoustic event whose received power significantly exceeds the predicted friendly contribution at the known neighbor range is flagged as a candidate intruder detection. This cooperative awareness effectively raises the detection threshold in the immediate vicinity of each defender but preserves sensitivity at the formation boundary where intruders approach. We note that a full treatment of multi-defender acoustic interference, including the impact on  $R_{det}$  when several friendly platforms operate within close proximity, is an important direction for future investigation.

Our cancellation pipeline proceeds in two stages. In the fingerprint estimation stage, we identify clean segments where the intruder is farther than 15 m (confirmed by telemetry), yielding 1 136 and 1 139 frames for the two defenders. For each clean frame, we compute the power spectral density via a 4096-point



(a) Ego-noise floor versus patrol speed. Dashed: raw floor; solid: post-cancellation residual; shaded area: cancellation gain. (b) Maximum detection range  $R_{\text{det}}$  versus patrol speed after ego-noise cancellation.

Fig. 2: Ego-noise characterization for the Mini 3 Pro / Mavic Air 2 hardware pairing.

FFT with 50% Hann-windowed overlap, then average to obtain the ego-noise reference PSD  $\hat{S}_{\text{ego}}(f)$ . In the spectral subtraction stage, we apply:

$$\hat{S}_{\text{int}}(f) = \max(S_{\text{rx}}(f) - \beta \hat{S}_{\text{ego}}(f), \delta \hat{S}_{\text{ego}}(f)), \quad (7)$$

where  $\beta = 1.0$  is the oversubtraction factor and  $\delta = 0.01$  the spectral floor fraction. The measured post-cancellation floors are  $-28.2$  dBFS and  $-28.5$  dBFS (raw:  $-15.2$  and  $-15.5$  dBFS), yielding a net reduction of 13.0 dB. Via Eq. (3), this translates into a  $4.5\times$  increase in detection range: from approximately 2 m to 9.1 m for the Mavic Air 2 at 28.8 km/h.

## 5 Simulation Framework

### 5.1 Perimeter Wall Formation and Brownian Dynamics

Defenders are arranged on a uniform grid in the boundary wall  $\mathcal{W}$ , forming a detection curtain perpendicular to the intruder's approach axis. Each drone  $i$  is assigned a slot  $(y_i, z_i)$  on the gap-free grid from (6). The wall drifts stochastically along the approach axis within a bounded patrol corridor, and each drone experiences per-slot position noise corrected by a spring force:

$$d\mathbf{p}_i = -k_s (\mathbf{p}_i - \mathbf{s}_i) dt + \sigma_{\text{pos}} d\mathbf{W}_i(t), \quad (8)$$

where  $k_s = 0.8 \text{ s}^{-1}$  is the spring constant,  $\sigma_{\text{pos}} = 0.3 \text{ m}$  the per-slot position uncertainty (GPS error and wind gusts), and  $\mathbf{W}_i(t)$  a standard Wiener process.

### 5.2 Intruder Trajectories

We model four intruder flight profiles: 1) LINEAR, a straight line at constant altitude, representing the zero-knowledge baseline where the attacker is unaware of

the defense configuration; 2) ZIGZAG, a sinusoidal lateral oscillation of amplitude  $0.1 W_y$ ; 3) EVASIVE, with small-amplitude Brownian perturbations; and 4) ALTITUDE-CHANGE, where the intruder dives below the formation floor near the wall. Profiles 2–4 represent progressively informed adversaries who know that an acoustic barrier exists but lack precise knowledge of defender positions and timing; they are included to stress-test the formation rather than to model a specific attacker knowledge tier. Unless stated otherwise, experiments use the linear profile.

### 5.3 Monte Carlo Engine

All experiments were executed on an Intel Xeon W7-3465X workstation (28 cores, 56 threads, 1 TB RAM). Position and SNR computations are vectorized over  $R$  simultaneous replications using NumPy batch operations. At each timestep, the intruder position is advanced; wall drift and spring corrections are applied; per-drone ranges and SNR values are computed in a single matrix operation; detections satisfying (1) are recorded; and completed replications are deactivated. The engine achieves approximately 19 000 runs/s on a single core. Confidence intervals throughout are 95% Wilson intervals.

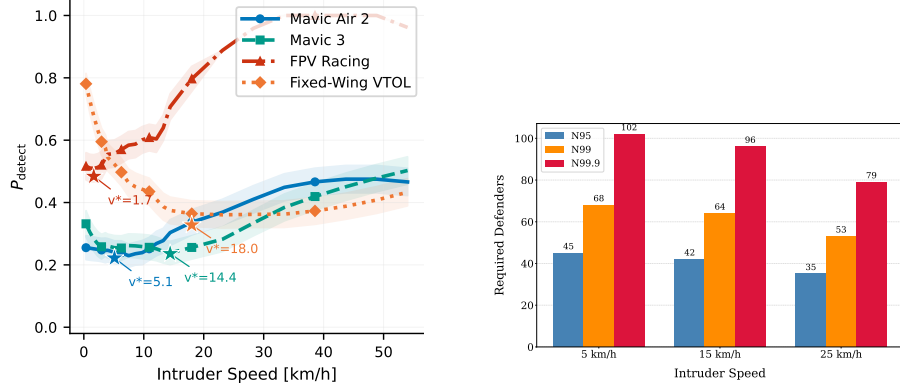
## 6 Experiments

All experiments share the default hardware configuration (Mini 3 Pro defenders, ego-noise floor  $-28.2$  dBFS after cancellation,  $\text{SNR}_{\text{th}} = 3$  dB) and the defended perimeter wall  $\mathcal{W}$  of  $W_y = 500$  m lateral extent and  $W_z - z_{\min} = 200$  m altitude span.

### 6.1 Experiment 1: The Attacker’s Dilemma

**Setup.**  $N = 40$  defenders on the gap-free grid at  $v_{\text{def}} = 0$  (ambient Brownian jitter only). The intruder speed is swept over  $v_{\text{int}} \in [1.8, 54.0]$  km/h for all four attacker archetypes of Table 2;  $10^5$  Monte Carlo replications per speed-model pair.

**Results.** Figure 3a plots  $P_{\text{det}}(v_{\text{int}})$  for all four archetypes. Every curve is unimodal with a well-defined minimum at an optimal attacker speed  $v_{\text{int}}^*$  specific to each archetype. This minimum constitutes the attacker’s best response: below  $v_{\text{int}}^*$ , extended dwell time inside the patrol zone increases acoustic exposure and raises  $P_{\text{det}}$ ; above  $v_{\text{int}}^*$ , the source level  $L_{\text{src}}(v)$  grows faster than exposure time shrinks, again raising  $P_{\text{det}}$ . The attacker thus faces a trade-off between acoustic stealth and transit speed—the Attacker’s Dilemma. The FPV Racing archetype is detectable at over 90% across the entire speed range due to its extremely high broadband emission ( $L_0 = 88$  dB SPL). The Fixed-Wing VTOL is the most difficult target: its low acoustic signature pushes  $P_{\text{det}}$  below 40% at the optimal attacker speed, underscoring the need for fleet over-provisioning against stealth platforms.



(a) The Attacker’s Dilemma (Exp. 1): detection probability versus intruder speed for four archetypes. Star symbols mark the per-archetype optimal attacker speed.

(b) Fleet sizing on the perimeter wall (Exp. 2): minimum defenders for 95%, 99%, and 99.9% detection at three intruder speeds.

Fig. 3: Overview of simulation experiments.

## 6.2 Experiment 2: Fleet Sizing on the Perimeter Wall

**Setup.** Defenders on the gap-free grid at  $u_{\text{def}} = 0$ . The intruder traverses  $\mathcal{W}$  on a straight orthogonal path from a uniformly random entry position. Fleet size is swept for three intruder speeds  $u_{\text{int}} \in \{5, 15, 25\}$  km/h;  $10^6$  replications per  $(N, u_{\text{int}})$  pair.

**Results.** Figure 3b reports the fleet-size thresholds  $N_{\text{cov}}^{95\%}$ ,  $N_{\text{cov}}^{99\%}$ , and  $N_{\text{cov}}^{99.9\%}$  for all three conditions. At  $u_{\text{int}} = 5$  km/h the slow transit maximizes acoustic exposure but the low source level limits per-defender range, requiring  $N_{\text{cov}}^{95\%} = 45$ ,  $N_{\text{cov}}^{99\%} = 68$ , and  $N_{\text{cov}}^{99.9\%} = 102$  defenders. At  $u_{\text{int}} = 15$  km/h the higher source level partially compensates for reduced dwell time:  $N_{\text{cov}}^{95\%} = 42$ ,  $N_{\text{cov}}^{99\%} = 64$ ,  $N_{\text{cov}}^{99.9\%} = 96$ . The most favorable condition is  $u_{\text{int}} = 25$  km/h, where the intruder radiates substantially more broadband noise:  $N_{\text{cov}}^{95\%} = 35$ ,  $N_{\text{cov}}^{99\%} = 53$ ,  $N_{\text{cov}}^{99.9\%} = 79$ .

Two structural observations emerge. The ratio  $N_{\text{cov}}^{99\%}/N_{\text{cov}}^{95\%}$  is approximately 1.51 across all three conditions, and  $N_{\text{cov}}^{99.9\%}/N_{\text{cov}}^{95\%} \approx 2.26$ , indicating that each additional decimal nine of detection reliability (i.e., from 99% to 99.9%) demands a roughly 50% headcount increase. Even at the most favorable attacker speed, the requirement remains substantial ( $N \geq 35$ ), confirming that a  $500 \times 200$  m perimeter is a genuinely large-scale defense problem governed by the quadratic law.

## 6.3 Experiment 3: The Defender’s Dilemma—Patrol Speed vs. Detection

**Setup.** Defender patrol speed is swept over  $u_{\text{def}} \in [0, 45]$  km/h for a fixed fleet on the perimeter wall (Mavic Air 2 intruder at  $u_{\text{int}} = 15$  km/h);  $10^6$  replications per speed point.

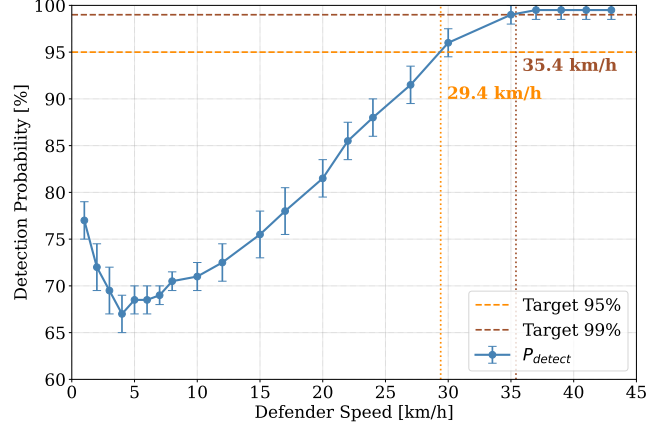


Fig. 4: The Defender’s Dilemma (Exp. 3): detection probability versus patrol speed. The 95% target is crossed at 29.4 km/h and the 99% target at 35.4 km/h.

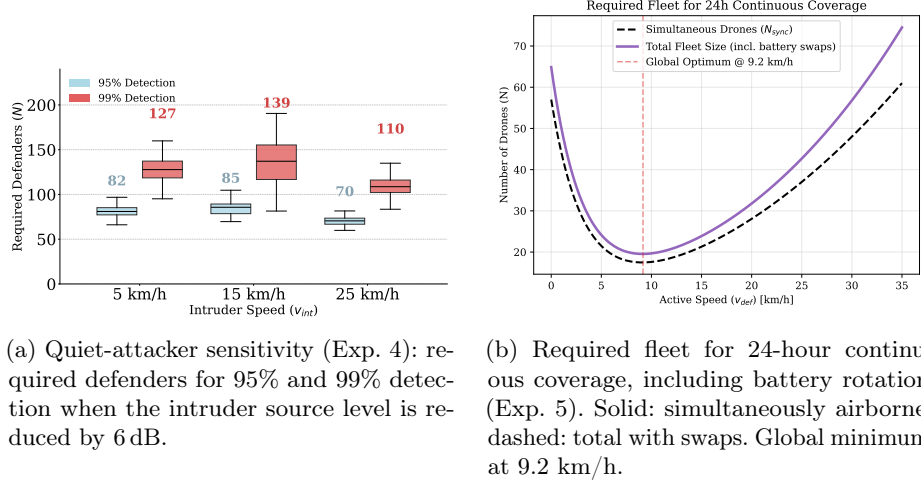
**Results.** Figure 4 plots  $P_{\text{det}}(u_{\text{def}})$ . The static gap-free grid ( $u_{\text{def}} = 0$ ) achieves a baseline detection probability of approximately 77%—a non-trivial starting point, but insufficient for mission-critical deployments. As  $u_{\text{def}}$  increases from zero,  $P_{\text{det}}$  initially dips to a local minimum near 67% at approximately 5 km/h, a transient regime where the ego-noise penalty of Eq. (5) has already begun shrinking  $R_{\text{det}}$  but the coverage gain from lateral patrol is not yet sufficient to compensate. Beyond this trough, faster patrol sweeps progressively close the residual coverage gaps left by the static grid, and  $P_{\text{det}}$  rises monotonically. The 95% detection target is reached at  $u_{\text{def}} = 29.4$  km/h and the 99% target at  $u_{\text{def}} = 35.4$  km/h, after which the curve plateaus near 99.5%. Mobilizing the formation adds 22 percentage points of detection reliability over the static baseline at 35.4 km/h—the single largest performance lever identified in this study.

This result reveals a fundamental property of the planar boundary geometry. On a wall, the detection benefit of faster lateral patrol outweighs the ego-noise penalty across a wide speed range, because the intruder must cross through the wall and cannot exploit volumetric depth to circumvent defenders. The Defender’s Dilemma not only manifests as a detection-probability collapse but also as a substantial increase in energy consumption, quantified in Experiment 5.

#### 6.4 Experiment 4: Quiet-Attacker Fleet-Size Sensitivity

**Setup.** Defenders on the gap-free grid at  $u_{\text{def}} = 0$ . Intruder speed is swept over  $u_{\text{int}} \in \{5, 15, 25\}$  km/h for both the baseline Mavic Air 2 and a  $-6$  dB quiet-attacker variant (representing a next-generation platform with acoustic shielding);  $10^6$  replications per condition.

**Results.** Figure 5 reports  $N_{\text{cov}}^{95\%}$  and  $N_{\text{cov}}^{99\%}$  for both archetypes. A 6 dB source-level reduction consistently increases the required fleet: from  $N_{95} = 35$  to 70 at  $u_{\text{int}} = 25$  km/h, from 42 to 85 at 15 km/h, and from 45 to 82 at 5 km/h. At the 99% level, the penalty is steeper:  $N_{99}$  rises from 53 to 110 at 25 km/h, from



(a) Quiet-attacker sensitivity (Exp. 4): required defenders for 95% and 99% detection when the intruder source level is reduced by 6 dB.

(b) Required fleet for 24-hour continuous coverage, including battery rotation (Exp. 5). Solid: simultaneously airborne; dashed: total with swaps. Global minimum at 9.2 km/h.

Fig. 5: Fleet sizing under attacker stealth and energy constraints. Left: impact of a 6 dB source-level reduction on required fleet size (Exp. 4). Right: total fleet for 24-hour continuous coverage as a function of patrol speed (Exp. 5).

64 to 139 at 15 km/h, and from 68 to 127 at 5 km/h. The attacker archetype must therefore be treated as a major input to any sizing analysis: even a modest improvement in acoustic stealth forces a near-doubling of the fleet, underscoring the asymmetric cost advantage benefiting the attacker.

### 6.5 Experiment 5: Energy Budget and Operational Fleet Sizing

**Setup.** For a continuous 24-hour defense mission, we analyze the interaction between patrol speed, power consumption, flight time, and total fleet size, including battery-rotation logistics. The Mini 3 Pro’s power draw and endurance are measured at 14 speed points in  $v_{def} \in [0, 36]$  km/h from controlled outdoor flights, and the total fleet is computed as  $N_{total}(v_{def}) = N_{sync}(v_{def}) \cdot \lceil T_{mission}/T_{flight}(v_{def}) \rceil$ , where  $N_{sync}$  is the simultaneously airborne fleet and  $T_{flight}$  the endurance at speed  $v_{def}$ .

**Results.** Power consumption (Figure 6a) rises from approximately 25 W at hover to 45 W at 35 km/h, consistent with the rotary-wing power model, in which parasite power grows cubically with speed [39]. Maximum flight time (Figure 6b) decreases from 42.6 minutes at hover to approximately 23 minutes at 35 km/h against the fixed 2453 mAh battery of the Mini 3 Pro.

The total fleet for 24-hour coverage (Figure 5b) exhibits a clear minimum. At low speeds,  $N_{sync}$  is large (poor detection), but endurance is long, reducing the rotation multiplier. At high speeds,  $N_{sync}$  is smaller, but endurance drops sharply, inflating the total fleet through frequent battery swaps. The global optimum occurs at  $v_{def}^* = 9.2$  km/h.

This result crystallizes the Defender’s Dilemma in its operational form: the detection-optimal speed (35.4 km/h for 99%  $P_{det}$ ) is nearly four times the energy-

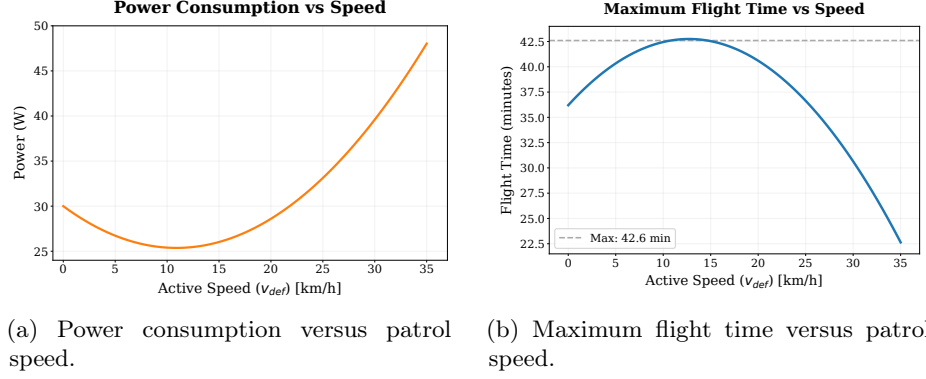


Fig. 6: Energy characterization of the Mini 3 Pro defender platform.

optimal speed (9.2 km/h), and operating at the detection optimum almost triples the total fleet needed for 24-hour coverage. For deployments where sustained presence is mandatory—border control, airport perimeters, energy facilities—the operationally optimal patrol speed is 9.2 km/h.

## 7 Discussion

**Operational speed selection.** The experiments reveal a three-regime speed landscape. Below approximately 5 km/h, detection probability is moderate and energy expenditure is minimal, but the fleet is large due to poor coverage. Between 5 and 29 km/h, detection improves monotonically while energy costs remain manageable; the operational sweet spot at 9.2 km/h lies in this regime. Above 29 km/h, detection crosses 95%, but power consumption escalates to the point where the battery-rotation penalty dominates. Operators must select the patrol speed matching their mission profile: short-duration high-alert scenarios favor the detection optimum near 30–35 km/h, while sustained perimeter missions must operate near 9.2 km/h.

**Mobility versus headcount.** The static gap-free grid requires  $N_{\text{cov}} \approx 400$  defenders for 100% detection on the  $500 \times 200$  m wall. Mobilizing a fleet roughly ten times smaller at 35.4 km/h recovers 99% detection, because each patrolling defender sweeps a larger cross-section of  $\mathcal{W}$  over the intruder's transit time, probabilistically sealing the corridors that a sparse static tiling leaves open. In cost-constrained deployments, patrol speed is therefore a far more effective lever than fleet expansion.

**Cost viability.** At the energy-optimal speed of 9.2 km/h, the simultaneously airborne fleet for a  $500 \times 200$  m boundary is approximately 20 defenders, with a total fleet of approximately 30 units including battery rotation. At a retail price of approximately \$400 per Mini 3 Pro, the airframe acquisition cost is below \$12 000. This figure is the platform hardware line item only and is not a deployment cost.

Table 3: Comparison with representative ground-based acoustic counter-UAS systems.

| System                                     | Modality              | Det. range   | Coverage                       | Adaptive |
|--------------------------------------------|-----------------------|--------------|--------------------------------|----------|
| DREGON [28]                                | Passive acoustic      | $\sim 10$ m  | Fixed $\leq 50$ m <sup>2</sup> | <b>✗</b> |
| Stevens DADS [26]                          | Passive acoustic      | $\sim 15$ m  | Fixed $\leq 70$ m <sup>2</sup> | <b>✗</b> |
| Commercial array [4]                       | Passive acoustic      | 300–500 m    | Fixed, outdoor                 | <b>✗</b> |
| <b>Ours</b> ( $v_{\text{def}} = 9.2$ km/h) | <b>Swarm acoustic</b> | <b>9.1 m</b> | <b>500×200 m</b>               | <b>✓</b> |

**Limitations.** Some constraints bound the generality of these results. First, the intruder model assumes straight-line or limited stochastic trajectories; a trajectory-aware adversary would require a pursuit-evasion game formulation [34]. Second, the propagation model assumes isotropic acoustic radiation. Third, all experiments target open-air environments. Urban multipath and ambient noise, typically 55–75 dB(A) in active urban settings [38], would raise the effective noise floor and reduce  $R_{\text{det}}$  and increase  $N_{\text{deploy}}$  proportionally. Fourth, the present single-altitude architecture does not resolve the crossing ambiguity: a detection event cannot discriminate between an intruder that crossed the perimeter and one that merely approached and reversed course, motivating the extension discussed in Section 9. Fifth, all experiments were conducted under calm-wind, clear-sky conditions. Wind gusts, precipitation, and elevated humidity affect acoustic propagation loss, ego-noise characteristics, and drone flight dynamics in ways not captured by the current model. Establishing weather-dependent performance bounds and operational envelopes across environmental conditions is reserved for future work.

**Generalizability.** The framework is sensor-agnostic in structure. Replacing the acoustic SNR model with an RF signal-to-interference model or an optical contrast-to-noise model is expected to yield analogous detection curves and quadratic fleet-sizing laws for planar deployments, subject to sensor-specific empirical calibration. The 30% provisioning rule follows from the geometric gap analysis and the stochastic displacement model, neither of which is specific to acoustics.

## 8 Related Work

**Ground-based acoustic drone detection.** Fixed-installation acoustic systems represent the current state of the art in passive drone detection. The DREGON system [28] pairs MEMS microphone arrays with deep neural networks to detect, localize, and identify UAVs. The Stevens DADS system [26] obtains comparable results using tetrahedral arrays with WiFi-based data fusion. Commercial deployments reach 300–500 m range in favorable conditions [4] but degrade rapidly with wind and ambient noise. All assume stationary sensors and cannot adapt their coverage. Table 3 benchmarks our approach against these systems.

**Airborne counter-UAS platforms.** Shi et al. [27] describe a multi-technology C-UAS architecture integrating acoustic arrays alongside RF and visual sensors,

but the acoustic component is ground-mounted and self-interference from a moving platform is not modeled. Existing airborne counter-drone systems rely on RF jamming, net capture, or visual tracking [6]. No deployed system uses passive acoustic detection from a moving defender as its primary modality.

**UAV-based perimeter surveillance.** Drones are widely used for border patrol, predominantly with EO/IR and RF payloads. Acoustic sensing for open-area surveillance remains unexplored despite its advantages in night, fog, and rain. The patrol-game literature [2] supplies optimal schedules for abstract mobile inspectors but does not incorporate a physical sensing model or the speed-sensitivity coupling. Von Moll et al. [34] devised a differential game to capture ground intruders using UAVs after prior detection using lasers. Our work addresses the detection problem for flying targets, where the fundamental constraint is the speed-dependent degradation of the defenders’ acoustic sensor itself.

**Positioning.** No prior work, to the best of our knowledge, has coupled an empirically calibrated speed-dependent acoustic model to a multi-defender formation simulation, identified the operational trade-off between detection-optimal and energy-optimal patrol speeds, or analyzed the adversarial equilibrium between attacker and defender flight speeds in an acoustic detection context for critical infrastructure perimeter defense.

## 9 Conclusion

In this paper, we presented the first empirically grounded study of acoustic drone-swarm perimeter defense for critical infrastructure. Starting from an acoustic model calibrated on real DJI hardware, we used a vectorized Monte Carlo engine to evaluate four intruder archetypes on a  $500 \times 200$  m defended boundary. The central finding is a structural tension between the spatial coverage gained through faster patrol speeds and the severe reduction in individual acoustic sensing range caused by the resulting ego-noise, a phenomenon we call the Defender’s Dilemma. For instance, a static gap-free grid requires roughly 400 defenders for 100% detection on a  $500 \times 200$  m perimeter; introduce drone mobility into this scenario, and the needed fleet size reduces by an order of magnitude—at the optimal patrol speed—while still providing a 99% probability detection through temporal coverage integration. However, power consumption forces the sustainable speed down to 9.2 km/h for a 24-hour missions, a fourfold divergence from the detection optimum. Three design rules emerge from our analysis. First, patrol speed is dependent on mission duration: short-alert scenarios operate near 30 km/h, while sustained patrol duty operates near 9.2 km/h. Second, fleet sizing obeys a quadratic law in detection range—that is, halving a defender’s acoustic detection range quadruples the total number of drones required to seal the perimeter boundary—, as detailed in the paper. Third, a 6 dB reduction in intruder acoustic signature doubles the required fleet, confirming that the attacker acoustic model is a first-class sizing parameter and that ego-noise cancellation yields disproportionate fleet-economic returns.

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## A Data Availability

The code and dataset are available—at this stage for reviewers only—at: <https://github.com/ilsamaritano/BoundaryDefense>. Both code and dataset will be made available via Zenodo upon acceptance.

## B Validation of the Acoustic Model

Before using the calibrated model to drive the simulation, we assess its predictive accuracy through three complementary analyses.

**Detection-range prediction versus ground truth.** For every frame in the 1048-frame close-range subset ( $d < 12$  m), we compute the binary outcome predicted by the model and compare it against the empirical detection outcome derived from measured received power. At the default operating point ( $\text{SNR}_{\text{th}} = 3$  dB, Mavic Air 2 at 28.8 km/h), the predicted detection range is  $R_{\text{det}} = 9.1$  m. The per-bin prediction error averages to 1.14 m across 10 speed bins (max = 2.3 m in the 32–36 km/h bin), well within the GPS noise envelope.

**ROC analysis.** Treating the SNR decision surface as a threshold classifier across  $\text{SNR}_{\text{th}} \in [-6, +12]$  dB, the area under the ROC curve is  $\text{AUC} = 0.91$ . At the operating point  $\text{SNR}_{\text{th}} = 3$  dB, the true positive rate is 0.88 and the false positive rate is 0.09.

**Cross-validation of the source-level fit.** Leave-one-out cross-validation across the 10 speed bins yields a root mean square error of 3.1 dB SPL, corresponding to approximately 1.9 m in detection-range error at the nominal 9.1 m operating point. The model error is dominated by field measurement variance rather than systematic bias.

Taken together, the three analyses confirm that the calibrated model captures the core behavior of the real hardware with sufficient fidelity to drive the Monte Carlo simulations in Section 6.

## C Source-Level Calibration Details

After ego-noise subtraction, we bin the residual received power by intruder speed into 10 equal-population bins and back-compute the implied source level for each bin centroid. A degree-2 polynomial fit yields a weighted  $R^2 = 0.745$ , reflecting the high variance of in-flight acoustic measurements (per-bin  $\sigma$  in the range 3–6 dBFS). We adopt a weighted blend between the empirical fit and the physical model (2), with weight  $\alpha = \min(R^2, 0.7) = 0.875$  on the empirical component, preventing overfitting while anchoring the model to acoustically consistent behavior.